

SHIVAM SINGH

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[LinkedIn](#) — [GitHub](#)

Summary

GenAI-focused developer with strong interest in **Agentic AI**, **LLM-based applications**, and **RAG pipelines**. Experienced in building AI-powered systems using **LangChain**, **Hugging Face**, **FastAPI**, and **Python**. Comfortable developing full-stack AI applications with **React.js**.

Education

B.Tech – Electronics & Communication Engineering **2023 – 2027**

Jalpaiguri Government Engineering College

Senior Secondary (Class XII) **2022**

Zoom International School

Core Interests

Generative AI, Agentic AI, Large Language Models, NLP, Retrieval-Augmented Generation, AI Automation

Technical Skills

Programming Languages: Python, C++

CS Fundamentals: Data Structures and Algorithms, DBMS, OOP, Machine Learning Fundamentals

GenAI / AI: LLMs, NLP, Transformers, Prompt Engineering, RAG, Embeddings, AI Agents, LangChain, LangGraph, LangSmith

Backend: FastAPI (REST APIs)

Frontend: HTML, CSS, JavaScript, React.js

Libraries & Tools: NumPy, Pandas, Scikit-learn, Hugging Face, Streamlit

Databases: MySQL

Version Control & DevOps: Git, GitHub, Docker

Projects

Agentic BlogWriter – Autonomous AI Blog Generation System *Python, LangGraph, LangChain*

- Built a multi-agent AI system that generates technical blog posts from a topic.
- Implemented router, research, planner and worker agents using LangGraph.
- Integrated web search and automatic citation generation.

HeartGPT – Emotional AI Chatbot *Python, FastAPI, Streamlit*

- Developed an AI-powered emotional chatbot for human-like conversations using LLMs.
- Implemented contextual responses with conversation memory.
- Built a responsive real-time chat interface.

YouTube Video RAG Chatbot *Python, LangChain, FAISS, Groq, Streamlit*

- Built a Retrieval-Augmented Generation system to answer questions from YouTube videos.
- Implemented transcript extraction and semantic search using vector embeddings.
- Integrated FAISS vector database and Groq LLM for fast contextual responses.
- Designed an interactive Streamlit interface for querying video content.

Embedding-Based Document Similarity System *Python, LangChain*

- Developed a semantic document similarity system using vector embeddings.

Movie Recommendation System *Python, Machine Learning*

- Implemented a content-based movie recommendation engine.